JOÃO RODRIGUES

PALO ALTO, CA | JOAOR.EU | CV@JOAOR.EU

Summary

I am a computational biologist with 15 years of experience and a proven track record leading research end to end across chemistry, biology, and computer science. My work has resulted in 46 peer-reviewed publications and several widely adopted open-source scientific software packages, totaling over 10,000 citations. Currently building workflows and agentic tools for structure prediction, cryo-EM structure refinement, and macrocycle modeling, while working directly with experimental groups to accelerate early-stage drug discovery efforts.

Experience

Schrödinger Inc | Senior Scientist → Senior Principal Scientist — 2021 - Present

- Lead a team of 5 scientists, engineers, and UI/UX researchers responsible for the Protein Preparation workflow, cryo-EM structure refinement, and macrocycle modeling products.
- Built evaluation pipelines and agentic tools on top of LLMs for Bunsen, an AI co-scientist, analyzing concrete failure cases to feed directly back into model-improvement workflows.
- Partnered with experimental biologists in the Structure Therapeutics Group to optimize drug discovery targets against low-resolution cryo-EM data, designing computational methods and pipelines to increase robustness and accuracy of subsequent screens.
- Designed and led a benchmark study on the real-world applicability of co-folding models for drug discovery, accepted for publication in npj Drug Discovery.
- Led scientific and engineering efforts on key macrocycle modeling products, resulting in improved scientific performance, a new docking method, renewed customer interest, several poster and oral presentations, and a publication in J. Med. Chem.
- Head the scientific development of the Protein Preparation Workflow, Schrödinger's most used tool (>250k executions per year), shipping improvements and new features with the UX/UI and scientific infrastructure teams.
- As an individual contributor, eliminated critical performance bottlenecks in flexible ligand docking approaches (IFD-MD) and implemented and validated new algorithms to extend their applicability to membrane proteins such as GPCRs.

Biopython Project | Maintainer and Core Developer — 2010 - Present

- Maintainer of the Bio.PDB module for structural bioinformatics.
- Implemented solvent-accessible surface area and structure alignment modules.
- Oversaw, reviewed, and shepherded individual contributions over the years.

Stanford University, Levitt Lab | Postdoctoral Fellow — 2016 - 2020

- Built a comparative modeling framework to predict cross-species transmission of SARS-CoV-2, integrating large-scale sequence, phylogeny, and structural data, resulting in 3 highly-cited publications and several national and international collaborations.
- Developed simulation methods to study the structure, dynamics, and interactions of membrane proteins, resulting in publications in Cell, Nat. Chem. Bio., and PNAS.
- Contributed several bug fixes and new features to OpenMM, a widely-used open-source engine for molecular dynamics simulations.
- Co-developed an immersive virtual reality approach to protein docking using Unity.
- Certified Software Carpentry instructor; taught Git, Python, and software best practices.

Utrecht University, Bonvin Lab | Doctoral and Postdoctoral Researcher — 2010 - 2016

- Core contributor to HADDOCK, a protein docking software with over 90,000 users across 190 countries, and consistently ranked among the best in CASP/CAPRI assessments.
- Co-developed coevolution-based methods based on large sequence datasets for protein quaternary structure prediction, a precursor to AlphaFold-based approaches.
- Created a widely-used open-source toolkit to manipulate atomic structures (pdb-tools).
- Created structural bioinformatics course materials.

Selected Publications

indicates corresponding author, * indicates co-first author

• Zhang, L, Friesner, RA, Miller, EB, and **Rodrigues, JP**# (2026) On the generalization and usability of cofolding models for GPCR drug discovery. npj Drug Discov.

• Robson-Tull, J, and **Rodrigues, JP**# (2026). Accurate Physics-Based Flexible Docking of Macrocyclic Ligands. J. Med. Chem.

• Miller, EB, Hwang, H, Shelley, M, Placzek, A, **Rodrigues, JP**, et al. (2024). Enabling structure-based drug discovery utilizing predicted models. Cell.

• Eastman, P, Galvelis, R, Peláez, RP, Abreu, CRA, Farr, SE, et al. (2024). OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. J. Phys. Chem. B.

• **Rodrigues, JP**#, Barrera-Vilarmau, S, M. C. Teixeira, J, Sorokina, M, Seckel, E, et al. (2020). Insights on cross-species transmission of SARS-CoV-2 from structural modeling. PLoS Comput Biol.

• Van Zundert, GCP, **Rodrigues, JP***, Trellet, M, Schmitz, C, Kastritis, PL, et al. (2016). The HADDOCK2.2 Web Server: User-Friendly Integrative Modeling of Biomolecular Complexes. Journal of Molecular Biology.

Selected Community Engagement

Invited Speaker, Pittsburgh Diffraction Conference 2026

Invited Panelist, Biophysical Society Meeting 2026

Invited Panelist, West Coast Structural Biology Workshop 2025

Invited Keynote, MICROBIOTEC'25, Azores (declined) 2025

Invited Panelist, Careers in Comp Chem, New York Area Group for Informatics and Modeling 2024

Scientific Committee, HIBIT Conference 2019-2025

Co-organizer, EMBO Practical Course on "Integrative Modeling of Biomolecular Interactions" 2016-2020

Invited Lecturer, Summer School "Exploring Nature's Molecular Machines", at Uni. Utrecht 2014

Co-Founder, Meeting for Portuguese researchers in computational biology (EJIBCE/3D-Bioinfo-PT) 2013-

Awards and Grants

The Molecular Sciences Software Institute (MOLSSI) - Seed Fellowship ($25,000) 2020

ISMB/ECCB - RCSB PDB Poster Prize - Outstanding Student Paper 2019

Niels-Stensen Fellowship for Postdoctoral Studies ($44,000) 2016

Stanford School of Medicine - **Dean's Postdoctoral Fellowship** ($25,000) 2016

Google Summer of Code Fellowship, Open Bioinformatics Foundation ($5,000) 2010

Education

Ph.D. in Computational Structural Biology 2010 - 2014
Bonvin Lab, Universiteit Utrecht, The Netherlands
Thesis: Computational Structural Biology of Macromolecular Interactions

M.Sc. in Biomolecular Sciences 2008 - 2010
Universiteit Utrecht, The Netherlands
Thesis: Predicting the Specificity of E2/E3 Complexes in the Ubiquitination Pathway

B.Sc. in Biochemistry 2004 - 2008
Universidade de Coimbra, Portugal
Thesis: Concept-based Query Expansion for Gene Association Prediction from Literature Data

---

Skills

Tech Stack Python, Cython, Bash, Git, SQL, Jira, CI/CD, Claude Code

Databases RCSB, NCBI, ChEMBL, BLAST, KEGG, Interpro, GPCRdb, OPM

Github github.com/joaorodrigues

Google Scholar scholar.google.com/citations?user=UeXRdzQAAAAJ